Physics 468/668: Statistical Physics **Spring 2019**

Lecture 20: Ising model in the zeroth approximation

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In 1928 Gorsky attempted a statistical study of order-disorder transitions in binary alloys on the basis of the assumption that the work expended in transferring an atom from an "ordered" position to a "disordered" one is directly proportional to the degree of order prevailing in the system. This idea was further developed by Bragg and Williams (1934, 1935) who, for the first time, introduced the concept of long-range order in the sense we understand it now and, with relatively simple mathematics, obtained results that could explain the main qualitative features of the relevant experimental data. The basic assumption in the Bragg-Williams approximation was that **the energy of an individual atom in the given system is determined by the (average) degree of order prevailing in the entire system rather than by the (fluctuating) configurations of the neighboring atoms**. In this sense, the approximation is equivalent to the mean molecular field (or the internal field) theory of Weiss, which was put forward in 1907 to explain the magnetic behavior of ferromagnetic materials.

20.1 Zeroth approximation

It seems natural to call this approximation the zeroth approximation, for its features are totally insensitive to the detailed structure. So the coordination numbers will be important.

We now define a long-range order parameter L in a given configuration by the very suggestive relationship

$$L = \frac{1}{N}\sum_i \sigma_i = \frac{N_+ - N_-}{N} = 2\frac{N_+}{N} - 1 \quad (-1 \le L \le +1) \tag{20.1}$$

which gives

$$\begin{aligned} N_+ &= \frac{N}{2}(1+L) \\ N_- &= \frac{N}{2}(1-L) \end{aligned} \tag{20.2}$$

The magnetization M is then given by

$$M = (N_+ - N_-)\mu - N\mu L \quad (-N\mu \le M \le +N\mu) \tag{20.3}$$

here L is a direct measure of the net magnetization in the system. For a completely random configuration, $\overline{N_+} = \overline{N_1} = \frac{1}{2}N$, the expectation values of L and M are zero.

The expression $-J(\frac{1}{2}q\bar{\sigma})\sum_i \sigma_i$ that is, For a given σ_i, we replace each of the $q\sigma_j$ by $\bar{\sigma}$ while the factor $1/2$ is included to avoid duplication in the counting of the nearest neighbor pairs, where q is the coordination number. Making use of equation (1), and noting that $\bar{\sigma} = \bar{L}$ we obtain for the total configurational energy of the system

$$E = \sum -J\sigma_i\sigma_j - (\mu B)NL = -J(\frac{1}{2}q\bar{\sigma})\sum_i \sigma_i - (\mu B)NL = -\frac{1}{2}(qj\bar{L})NL - (\mu B)NL \tag{20.4}$$

The expectation value of E is

$$U = -\frac{1}{2}qJN\bar{L}^2 - \mu BN\bar{L} \tag{20.5}$$

In the same approximation, the difference $\delta\epsilon$ between the overall configurational energy

$$\begin{aligned} \delta\epsilon &= -J(q\bar{\sigma})\delta\sigma - \mu B\delta\sigma \\ &= 2\mu\left(\frac{qJ}{\mu}\bar{\sigma} + B\right) \end{aligned} \tag{20.6}$$

For here $\Delta\sigma = -2$. The quantity $(qJ\bar{\sigma}/\mu)$ thus stands for the internal field, which is determined by (i) long range ordering (ii) the coupling strength. The relative values of the equilibrium numbers $\bar{N_+}$ and $\bar{N_-}$ then following Boltzmann principle.

$$\frac{\overline{N_-}}{\overline{N_+}} = \exp(-\delta\epsilon/kT) = \exp[-2\mu(B' + B)/kT)] \tag{20.7}$$

Where B' denotes the internal molecular field.

$$B' = qJ\bar{\sigma}/\mu = qJ(\bar{M}/N\mu^2) \tag{20.8}$$

Therefore,

$$\frac{1-\bar{L}}{1+\bar{L}} = \exp[-2(qJ\bar{L} + \mu B)/kT] \tag{20.9}$$

or

$$\frac{qJ\bar{L} + \mu B}{kT} = \frac{1}{2}\ln\frac{1-\bar{L}}{1+\bar{L}} = \tanh^{-1}\bar{L} \tag{20.10}$$

To investigate the possibility of spontaneous magnetization, we let B $\to 0$, which leads to the relationship

$$\overline{L_0} = \tanh\left(\frac{qJ\overline{L_0}}{kT}\right) \tag{20.11}$$

This equation can be solved graphically, For any temperature T, the appropriate value of $\overline{L_0(T)}$ is determined by the point of intersection of (i) the straight line $y = L_0$ and (ii) the curve $y = \tanh(qJL_0/kT)$. Except $\overline{L_0} = 0$, we look for the nonzero solution.

Graphically, it can be seen that nonzero solution only exists when

$$qJ/kT > 1 \to T < qJ/k = T_c \tag{20.12}$$

We thus obtain a critical temperature T_c, below which the system can acquire a nonzero spontaneous magnetization and above which it cannot. Also, we see that if L_0 is a solution, $-L_0$ is the solution as well.

The precise variation of $\overline{L_0}(T)$ with T can be solved numerically. If we use the approximation $\tanh x \approx x - x^3/3$, we get

$$\overline{L_0}(T) \approx [3(1 - T/T_c)]^{1/2} \tag{20.13}$$

On the other hand, as $T \to 0$, $\overline{L_0} \to 1$ from eq. 20.11,

$$\overline{L_0}(T) \approx 1 - 2\exp(-2T_c/T) \quad (T/T_c \ll 1) \tag{20.14}$$

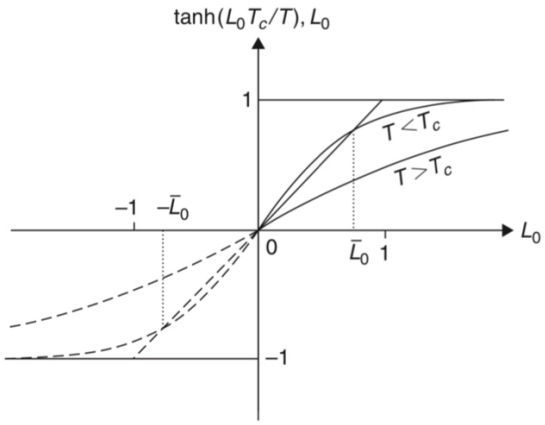


Figure 20.1: The graphical solution

The energy and specific heat are

$$U_0(T) = -\frac{1}{2}qJN\overline{L_0}^2 \tag{20.15}$$

and

$$C_0(T) = -qJN\overline{L_0}\frac{d\overline{L_0}}{dT} = \frac{Nk\overline{L_0}^2}{(T/T_c)^2/(1-\overline{L_0}^2) - T/T_c} \tag{20.16}$$

Thus, for all $T > T_c$, both $U_0(T)$ and $C_0(T)$ are zero. While at the high temperature limit ($x = TcL_0/T$),

$$C_0(T_c) = \lim_{x\to 0}\left\{\frac{Nk\cdot 3x}{\frac{(1-x)^2}{1-3x} - (1-x)}\right\} = \frac{2}{3}Nk \tag{20.17}$$

$$\begin{aligned} S_0(T_c) &= \int_0^{T_c}\frac{C_0(T)dT}{T} = -qJN\int_1^0 \frac{\overline{L_0}}{T}d\overline{L_0} \\ &= Nk\int_0^1 (\tanh^{-1}\overline{L_0})d\overline{L_0} = Nk\ln 2 \end{aligned} \tag{20.18}$$

$$\chi(B,T) = \left(\frac{\partial\bar{M}}{\partial B}\right)_T = N\mu\left(\frac{\partial\bar{L}}{\partial B}\right)_T = \frac{N\mu^2}{k}\frac{1-\bar{L}^2(B,T)}{T - T_c[1-\bar{L}^2(B,T)]} \tag{20.19}$$

For $\bar{L} \ll 1$ (which is true at high temperatures for a wide range of B but is also true near T_c if *B is small*), we obtain the Curie-Weiss law,

$$\chi_0(T) \approx (N\mu^2/k)(T - T_c)^{-1} \tag{20.20}$$

For T less than, but close to, T_c we utilize equation 20.14 as well and get

$$\chi_0(T) \approx (N\mu^2/2k)(T_c - T)^{-1} \tag{20.21}$$

Finally, using the approximation of

20.2 Remarks

Clearly there is similarity that exists between the critical behavior of a gasliquid system obeying van der Waals equation of state and that of a magnetic system treated in the Bragg-Williams approximation. Even though the two systems are physically very different, the level of approximation is such that the exponents governing power-law behavior of the various physical quantities in the critical region turn out to be the same; compare, for instance, equation (14) with (12.2.11), equations (22a) and (22b) with (12.2.13) and (12.2.14), equation (24) with (12.2.8) along with the behavior of the specific heat as well. This sort of similarity will be seen again and again whenever we employ an approach similar in spirit to the mean field approach of this section.

20.3 Random mixing approximation

The mean configurational energy in the absence of the external field is given by

$$U_0 = -J\left(\frac{1}{2}qN - 2q\overline{N}_+ 4\overline{N}_{++}\right) \tag{20.22}$$

At the same time,

$$\overline{N}_+ = \frac{1}{2}N(1+\bar{L_0}), \quad U_0 = -\frac{1}{2}qJN\bar{L_0}^2 \tag{20.23}$$

Therefore, we have

$$\bar{N}_{++} = \frac{1}{8}qN(1+\bar{L_0})^2 \tag{20.24}$$

so that

$$\frac{\bar{N}_{++}}{\frac{1}{2}qN} = \left(\frac{\bar{N}_+}{N}\right)^2 \tag{20.25}$$

Thus, the probability of having an up-up nearest neighbour pair of spins in the lattice is precisely equal to the square of the probability of having an up spin. In other words, short range ordering does not have an impact on the results.

$$\bar{N}_{++} = \frac{1}{2}qN\left(\frac{1+\bar{L}}{2}\right)^2, \quad \bar{N}_{--} = \frac{1}{2}qN\left(\frac{1-\bar{L}}{2}\right)^2 \tag{20.26}$$

$$\bar{N}_{+-} = 2*\frac{1}{2}qN\left(\frac{1+\bar{L}}{2}\right)\left(\frac{1-\bar{L}}{2}\right) \tag{20.27}$$

with the result that

$$\frac{\bar{N}_{++}\bar{N}_{--}}{(\bar{N}_{+-})^2} = \frac{1}{4} \tag{20.28}$$